

Chapter 7

Oil Product Markets

What a barrel becomes — and how refined fuels are priced and traded

One barrel, many products

42

gallons in a barrel

Each 42-gallon barrel of crude is split into a slate of refined products — gasoline, diesel, jet fuel, fuel oil and more. Crude is only the raw material; these fuels are what the world actually buys, and each trades in a market of its own.

A barrel is a bundle — No one consumes crude oil directly. Its value comes entirely from the products refined out of it.

Products are what we buy — The fuel at the pump, in the jet, and in the truck is a refined product — priced separately from the crude it came from.

Two prices, one barrel — Crude is the input cost; products are the revenue. The gap between them decides whether a refinery runs at all.

From crude to products

Distillation first — Crude is heated in a tower and separated by boiling point: light gases at the top, gasoline and naphtha in the middle, diesel and gasoil below, heavy residue at the bottom.

Light, middle, heavy — These cuts group into product families. The 'middle distillates' — diesel and jet — sit at the heart of industrial and transport demand.

Conversion adds value — Complex refineries use cokers and crackers to turn cheap heavy residue into valuable light products, lifting the worth of every barrel.

Yield depends on the crude — A light, sweet barrel yields more fuel with less processing; a heavy, sour barrel needs far more work to reach the same slate.

The product slate

Gasoline — The single largest product, fueling the world's cars. Demand peaks during the northern-hemisphere summer driving season.

Diesel & gasoil — The workhorse fuel of trucks, trains, farms, and industry. Its demand tracks the real economy more closely than any other product.

Jet fuel (kerosene) — Tied to air travel and closely related to diesel; among the products most sensitive to the health of the global economy.

Fuel oil, LPG & naphtha — Heavy fuel oil powers ships; LPG heats and cooks; naphtha feeds the petrochemical industry that makes plastics.

The crack spread — a refiner's margin

The core idea — A refiner's profit is the gap between what it pays for crude and what it earns selling products. That gap is the crack spread.

The 3-2-1 — The standard benchmark assumes three barrels of crude yield two of gasoline and one of distillate. It is the most-watched gauge of refining health.

Why it matters — A wide crack pushes refiners to run flat out, pulling more crude through the system; a thin or negative crack forces run cuts and tightens product supply.

It can be traded — Refiners hedge their margin with crack-spread futures on the CME, locking in profit and passing price risk to the financial market.

Gasoline — the largest market

The benchmark — In the United States, the futures contract is RBOB gasoline, traded on NYMEX and quoted in cents per gallon.

Seasonality rules — Demand and price climb into the summer driving season and ease in winter, a rhythm reinforced by seasonal fuel-blend requirements.

Octane and blending — Gasoline is a blend tuned to octane and emissions specs that vary by region and season, creating distinct local markets.

Sensitive to the consumer — As the fuel people buy at the pump, gasoline prices feed straight into household budgets and into politics.

Diesel — the economy's fuel

The industrial barometer — Diesel and gasoil move freight, run machinery, and power generators. When the economy accelerates, distillate demand leads the way.

Structurally tight — Diesel is harder to make and harder to substitute than gasoline, so its market can squeeze sharply when supply is disrupted.

Global, not local — Diesel trades across oceans between regions, so a shortage in one place is quickly felt everywhere.

The 2022 lesson — After Europe lost Russian diesel, cracks spiked to a record near \$83 a barrel — a reminder of how violently this market can tighten.

The lighter and heavier ends

Jet fuel — A close cousin of diesel, priced off the same middle-distillate pool. Its demand swings with air travel and the travel season.

Fuel oil — The heavy residue at the bottom of the barrel, burned by ships and some power plants. Low-margin, but part of every barrel's slate.

The IMO 2020 shift — Tighter marine-fuel sulfur limits pushed shipping toward cleaner fuels, permanently widening the value gap between light and heavy products.

LPG & naphtha — The lightest cuts feed heating, cooking, and the petrochemical plants that turn oil into plastics and chemicals.

Benchmarks and trading hubs

United States — RBOB gasoline and ULSD (heating oil) futures on NYMEX, with the US Gulf Coast as the physical pricing center.

Europe — ICE low-sulfur gasoil is the diesel benchmark, anchored on the Northwest Europe (ARA) refining and storage hub.

Asia — Singapore is the pricing hub for the fastest-growing product market, setting the marker for the entire region.

Arbitrage links them — When one region's product price runs ahead, cargoes flow in to capture the spread, tying the global market together.

What moves product prices

Crude, with a lag — Products follow crude, but not instantly. After a crude spike, refiners selling at new prices can briefly capture a wider margin.

Refining capacity — Unplanned outages, run cuts, and scheduled maintenance turnarounds in spring and autumn tighten supply and lift cracks.

Seasonality — Gasoline strengthens through summer, distillates through winter. Each product runs on its own calendar.

Regulation and quality — Sulfur limits, blend mandates, and biofuel rules reshape which products are valuable and where they can be sold.

The product market today

Spike and lag — When Brent surged in early 2026, product prices followed with a delay, briefly handing refiners unusually wide margins.

Cracks blew out — With Gulf supply disrupted, the gasoline crack ran toward \$28 a barrel and distillate cracks tightened sharply as scarce barrels were rationed by price.

The pump felt it — US retail gasoline reached about \$3.99 a gallon and diesel about \$5.40 at the end of March 2026, feeding straight into inflation.

Easing into mid-year — As crude retreated below \$100 and Hormuz traffic began to recover, product prices and cracks started to normalize.

Conclusion — oil meets the consumer

Products are the point — Crude is only valuable for what it becomes. The product markets are where oil finally meets the consumer.

The crack is the link — The spread between crude and products governs how hard refiners run and, ultimately, how much fuel reaches the market.

Each product is its own market — Gasoline, diesel, and jet each have distinct demand, seasonality, and benchmarks, yet all move together with crude.

What to remember — To read the oil market fully, watch not just the price of crude but the spreads to the fuels it becomes — that is where refining profit and real demand are decided.